

An investment fund is trying to determine how to invest its assets for the following year, in order to maximize its profit. Currently, the fund has 2.5 million euros that it can invest in state bonds, real estate loans, car loans or scholarship loans. The annual interest rates of the listed investment types are 4% for state bonds, 6% for real estate loans, 8% for car loans and 9% for scholarship loans.

To minimize risks, the investment fund allows only the selection of a strategy satisfying the following restrictions:

- the amount invested in car and scholarship loans must not exceed twice the amount invested in bonds;
- the amount invested in car loans must be larger or equal than the amount invested in scholarship loans;
- the investment in car loans should not exceed the investment in real estate loans by more than 70%.

Formulate this problem as an optimization problem by determining

1. the decision variables,
2. the objective function, and
3. the constraint(s).